

Emotional Expressivity Biomarkers for PTSD, Depression, and Anxiety Detection

Executive summary

Emotional expressivity features are promising **digital behavioral biomarkers** for psychiatric screening and monitoring, but the evidence is much stronger for **adjunctive assessment** than for stand-alone diagnosis. Across the studies reviewed here, the most reproducible signals are not generic “emotion scores” alone, but **interpretable low-level features**: facial action units (AUs), head-motion dynamics, gaze/eye-region behavior, and temporal variability of these signals over time. In depression, the literature most consistently points to **reduced positive expressivity, altered brow and mouth activity, and reduced movement variability/flexibility**. In PTSD, the strongest evidence is **context-sensitive and multimodal**: visual arousal markers, facial movement parameters, and speech/language cues jointly outperform any single channel, and recent child-PTSD work specifically emphasizes **de-identified AU, landmark, gaze, and head-pose sequences**. In anxiety, the signal appears comparatively less “blunted” and more **tension/arousal-oriented**, with **anger/fear/neutral expression differences, head rotation, jawline/eye-region motion, and gaze-related cues** standing out. ¹

The provided repository is a solid **interpretable baseline**. Its current emotional expressivity path is centered on **py-feat-derived basic emotion probabilities, AUs, and mouth openness**, with optional baseline normalization and a speaking/non-speaking split. Those are useful and clinically sensible starting points. However, compared with the stronger literature, the current implementation under-emphasizes **explicit gaze/head-pose outputs, temporal dynamics beyond mean and standard deviation, context-aware modeling, multimodal fusion, uncertainty estimation, and robust external-validation protocols**. The code already contains pieces that are directionally aligned with the literature—such as baseline normalization and speech-aware splitting—but the strongest next step is to evolve from a framewise summary toolbox into a **windowed, sequenced, multimodal digital biomarker pipeline**. ²

A practical recommendation for this repo is to keep the current AU/emotion summary as the **transparent baseline**, then add three layers on top of it: **quality-controlled feature extraction** (AUs, head pose, gaze, blink, asymmetry, confidence flags), **temporal summarization and sequence modeling** (2–10 second windows with overlap, entropy/autocorrelation/onset-offset statistics, transformer or BiLSTM baselines), and **multimodal fusion** with audio text/ASR and optional physiological features. The best evidence in the reviewed literature supports this progression much more strongly than a move toward opaque, raw-video-only models. ³

Review of the repository and current measurement logic

The inspected repository material shows two separate but related design ideas. The markdown at `emotional_expressivity/emotional_expressivity.md` describes emotional expressivity as a framewise quantification of **happiness, sadness, anger, fear, disgust, surprise, and neutral**, with

summary statistics and an optional split into **speaking** and **not speaking** periods. It also recommends cautious use in settings with **occlusion, poor video quality, multiple faces, or poor lighting**, and positions a baseline video as an optional subject-specific normalization reference. ⁴

The repo structure and README diagrams indicate that the broader `openwillis-face` stack is organized around preprocessing plus specialized modules such as `face_landmark.py`, `eye_blink.py`, and `facial_emotion.py`, with recent OpenWillis release notes explicitly noting improved preprocessing for **multiple-face videos** to support downstream facial and emotional expressivity measurement. That architecture is clinically sensible: preprocessing and track stabilization first, phenotype extraction second, summarization last. ⁵

The core implementation in `facial_emotion.py` is straightforward and interpretable. It uses a shared **py-feat Detector**, extracts **emotion probabilities** and **AUs** framewise, appends a custom **mouth_openness** variable derived from lip-landmark distances, and samples the video according to `skip_frames` with a default of 5, which effectively makes the expressivity pass much sparser than the original frame rate. If a baseline video is provided, the code normalizes framewise outputs by the baseline mean after a `+1` offset. This is transparent and easy to reason about, but it also means the shipped summary is currently built around **mean-level relative activity**, not around richer dynamics such as burstiness, persistence, or transition structure. ⁶

The speaking split is implemented in `speaking_utils.py` using the **rolling standard deviation of mouth_openness**, standardized and separated by a **2-component Gaussian mixture model** into likely speaking vs non-speaking states, after which the summary is reduced mainly to **mean, standard deviation**, and an `overall_expressivity` aggregate. This is clever as a low-cost approximation, but it is also a likely source of confounding because mouth movement reflects not only speech, but also smiling, tension, lip pressing, breathing, and task demands. In other words, the repo already contains the right instinct—speech-context control—but currently does it with a proxy that the reviewed literature suggests should ideally be paired with **audio voice activity detection and active-speaker logic**. ⁷

One code-level issue is worth flagging for maintenance: the helper `bb_dict_to_bb_list` appears to construct the lower y-coordinate as `bb_y + bb_y` rather than `bb_y + bb_h`, which looks like a likely bounding-box bug rather than an intended transform. Even if that helper is not central to the main path, it is the kind of silent geometry error that can degrade crop quality in exactly the kinds of subtle-expression problems this project targets. ⁶

Evidence across disorders

The table below prioritizes primary studies and a small number of high-value preprints. Where the accessible source exposed only an abstract or snippet, that limitation is shown as **NR** rather than inferred.

Disorder	Study	Dataset / setting	Population	Features	Measurement and computation details	Models	F
Depression	Alghowinem et al., 2013 ⁸	Clinically validated depression videos	Adults; depressed vs non-depressed; exact N not recoverable from inspected source	Head pose and movement	3D face model projected onto 2D AAM; low-level + statistical functionals; leave-one-subject-out CV	SVM	F r n a r s n k c s v n f
Depression	Harati et al., 2020 ⁹	Repeated clinical interviews during DBS recovery	12 severely depressed patients, longitudinal	Dynamic facial variability from muted video	Multiscale entropy (MSE) and temporal variability features from repeated interviews	Ordinal regression with elastic-net regularization	H e v t c s P i n b e c j i b A s e c t

Disorder	Study	Dataset / setting	Population	Features	Measurement and computation details	Models	Performance
Depression	Jiang et al., 2021 ¹⁰	365 video interviews, 88 hours, DBS cohort	12 depressed patients, repeated over time	7 facial emotions, AUs , temporal statistics	Regional CNN detector + ImageNet-pretrained CNN for basic emotions; OpenFace for AUs; temporal statistics over recordings	Classification of remission / treatment response	A r c C t r l s v s c g l f v
Depression	Mahayossanunt et al., 2023 ¹²	Clinical interview videos collected at Chulalongkorn University	474 videos; 134 depressed / 340 non-depressed	Gaze angles, angular/radian features, AU intensity , facial-expression features	Privacy-protected structured facial features from interview video; detailed frame-rate/windowing not exposed in inspected snippet	LSTM with attention-based intermediate fusion + label smoothing	A 9 8 P 9 8 i g t p f i

Disorder	Study	Dataset / setting	Population	Features	Measurement and computation details	Models	F
Depression	Kim et al., 2023 ¹³	Controlled posed and spontaneous emotion tasks	59 older adults without cognitive impairment	17 AUs across six emotions; posed vs spontaneous patterns	OpenFace 2.0 / FACS-style AU extraction; PCA + multiple regression	Association study, not classifier	F c s l c i t c e n r k s e s s e s
Depression	Jin et al., 2025 ¹⁵	E-DAIC clinical interview corpus	219 participants; train 163 / val 56 / test 10	OpenFace facial features, gaze, head pose, AUs plus audio MFCC	E-DAIC provides extracted privacy-preserving features; depression defined as PHQ-8 > 10 ; every 100 frames aggregated for explanation	TSNet-DD for video, GCN-LSTM for audio, VAFN multimodal fusion	V C C n f C C T e a A a c

Disorder	Study	Dataset / setting	Population	Features	Measurement and computation details	Models	F
PTSD + Depression	SchulteBraucks et al., 2022 ¹⁷	Trauma-survivor free-speech interviews, 1 month post-ED	81 trauma survivors	Facial emotion/ intensity, movement parameters , speech prosody, language	Deep neural network extracted visual, acoustic, and semantic markers from open-ended qualitative interviews	Cross-sectional classification	F
PTSD	Aathreya et al., 2025 ¹⁸	Child PTSD dataset across seven conversational contexts	18 children	AU intensities, facial landmarks, eye gaze, head pose	De-identified temporal features extracted per video; seven different contexts per session	Transformer with learnable Fourier encoding on AU sequences	F

Disorder	Study	Dataset / setting	Population	Features	Measurement and computation details	Models	F
PTSD	Fujiwara et al., 2015 ¹⁹	Emotional-reactivity pilot after disaster exposure	23 preschool children exposed to the Great East Japan Earthquake	Facial-expression ratings to a comedy clip	Behavioral/emotional reactivity paradigm, not automated CV classification	Association / pilot study	I k b s i e n e n t c b C a n l c h
Anxiety	Ren et al., 2025 ²⁰	IAPS-based facial-expression elicitation for GAD screening	60 GAD patients + 60 matched controls	Seven facial-expression categories; especially neutral, anger, fear	Facial expression recognition in an aided diagnostic paradigm; validity against GAD-7 and HAMA ; diagnosis benchmarked against MINI	ROC-based screening analysis	E s c n - A a f n s n t t a

Disorder	Study	Dataset / setting	Population	Features	Measurement and computation details	Models	F
Anxiety / symptom triad	Zhou et al., 2023 21	MCI outpatient recordings with speech/ facial capture	319 older adults with mild cognitive impairment	Speech, facial- expression, and text features	Open-source toolkits used for speech, facial, and text extraction	Random forest / multiclass symptom models	T s a a n c a c a p a p t n s i i P s a

Disorder	Study	Dataset / setting	Population	Features	Measurement and computation details	Models	F
Anxiety-like social state	AnxietyFaceTrack, 2025 preprint ²²	Smartphone videos during 2-minute idle exposure with unfamiliar peers	91 students; 1,173 10-second samples	OpenFace-derived 669 features from landmarks, eye gaze, pose, AUs	30 fps , non-overlapping 10 s windows , mean pooling per window; OpenFace generated 714 features and 669 were retained	Random Forest best	M a F C a C n f e v i p n r a e f n T b a s

The broad pattern from these studies is clear. **Depression** has the deepest facial-computing literature and repeatedly benefits from **AUs plus dynamics plus multimodal fusion**. **PTSD** has fewer facial-only studies, but the strongest work shows high value for **post-trauma free speech** and **context-aware multimodal markers**. **Anxiety** appears more heterogeneous: stronger results often arise either in **specific diagnostic paradigms** such as GAD screening or in **state-anxiety/social-threat setups**, where **head pose and eye-region cues** can be as important as, or more important than, coarse emotion classes. ²⁴

Biomarker patterns, disorder contrasts, and confounds

A useful way to read this literature is to separate **what is consistently visible** from **what is likely transdiagnostic**. Most studies do **not** support a simple mapping such as “one disorder = one facial expression.” Instead, the evidence points to **families of expressivity markers** whose diagnostic value depends on context, time scale, conversation type, and comorbid symptoms. ²⁵

Biomarker synthesis by feature family

Feature family	Depression	PTSD	Anxiety disorders / anxious states	Robustness and caveats
Reduced positive expressivity / smile attenuation	Most consistent depression signal; related to social withdrawal and diminished positive affect; older and prior depression work particularly implicate mouth-corner and smile-related changes. ²⁶	Can appear, but PTSD evidence is more context-sensitive and less consistently “blunted” than depression. ²⁷	Less central than in depression; anxiety is more often signaled by tension, fear, vigilance, or social discomfort than by simple smile reduction. ²⁸	Strongly confounded by interview topic, speaking load, and sociability.
Brow tension / inner-brow elevation / narrowing	Present in older-adult depression findings and echoed in recent subthreshold-depression work. ²⁹	Likely relevant through arousal/threat processing, but not yet well standardized as a stand-alone PTSD biomarker. ³⁰	Fits GAD/threat-tension paradigms well, especially when fear/anger are modeled. ²⁰	Likely transdiagnostic rather than disorder-specific.
AU intensity patterns	Very strong evidence. In E-DAIC explanations, AU contributions exceeded pose and gaze; interview-based LSTM/attention models using AUs perform well. ³¹	Strong recent direction. Child-PTSD work explicitly chose AU intensities as the best baseline feature family. ³²	Useful, but not always dominant; anxiety models often need pose/eye-region cues too. ²⁸	Probably the best single facial feature family for the repo to prioritize.

Feature family	Depression	PTSD	Anxiety disorders / anxious states	Robustness and caveats
Temporal variability / entropy / flexibility	One of the most clinically meaningful depression markers; reduced variability tracks greater severity, while recovery studies show informative changes over time. ⁹	Emerging evidence suggests day-to-day emotion-expression dynamics may reveal PTSD maintenance mechanisms. ³³	Anxiety may show more rapid tension fluctuations and context-linked instability than tonic flattening. ³⁴	Underreported in many benchmark papers; should be added to the repo.
Head pose / head movement	Strong and long-running cue family; even early head-motion-only depression models performed reasonably well. ⁸	Included in both trauma-survivor multimodal work and child-PTSD datasets, but seldom isolated as the main signal. ³⁵	Particularly salient for anxiety-like states; head rotation was among the most important features in AnxietyFaceTrack, and pose-only features remained strong. ³⁶	High-value feature family missing from the repo's current emotional summary output.
Eye gaze / blink / eye-region cues	Usually helpful but often weaker than AUs plus audio in depression fusion models. ³⁷	Included in modern PTSD datasets; likely especially useful for avoidance/vigilance patterns, but evidence is still comparatively sparse. ³⁸	Clinically plausible and repeatedly discussed; eye-gaze-only performance can be modest, but the eye region still contributes. ²⁸	Best used jointly with AUs and head pose rather than alone.
Speech-linked mouth movement	Important confound. Free-speech depression interviews are useful, but uncorrected mouth activity can reflect speaking rate more than affect. ³⁹	Also crucial in PTSD post-trauma interviews, where arousal is expressed during narrative recall. ⁴⁰	Can mimic or mask anxiety-related tension, especially in interview tasks. ⁴¹	The repo is right to split speaking vs non-speaking, but its current proxy should be upgraded.

Only a minority of studies reported **standardized effect sizes** in a way that makes biomarker interpretation easy. The clearest example in the retrieved set is the GAD validation study, where expression-symptom correlations ranged from about **-0.35 to 0.34**, which is a small-to-moderate range rather than a large one. Most other studies report **AUC, F1, MAE, or accuracy**, which is useful for prediction benchmarking but much less useful for deciding whether a feature is robust enough to claim as a disorder-specific biomarker. That gap is an important weakness of the field. ⁴²

The main confounds are also consistent across the literature. **Medication and treatment state** can change expressivity, as shown in DBS recovery work and in studies where current treatment status is acknowledged as a likely source of signal distortion. **Comorbidity** is substantial: post-trauma depression overlaps with PTSD, and older-adult MCI work often models depression, anxiety, and apathy jointly. **Demographics** matter too: both Jin et al. and AnxietyFaceTrack observed performance differences by sex, and age-dependent effects appear in both older-adult and kinematic work. These are not side notes; they affect which features seem “important.” ⁴³

Methodological quality and limitations

The field is promising, but methodologically uneven. The strongest papers use **clinical samples, subject-exclusive validation**, and clearly defined outcomes. Even then, many datasets are small. The DBS facial studies rely on **12 patients**, the trauma-survivor PTSD/depression classifier uses **81 participants**, the child-PTSD dataset contains **18 children**, the older-adult depression study has **59 participants**, and the GAD validation study includes **120 total subjects**. Only some depression benchmarks, such as E-DAIC and the 474-video Chulalongkorn set, reach sizes that begin to support more stable model comparison. ⁴⁴

A second problem is **label heterogeneity**. Some studies model **clinician-anchored diagnosis or validated clinical scales** such as MINI, PHQ-8, GAD-7/HAMA, or PTSD status. Others rely on broader symptom questionnaires such as DASS or even a **single self-report state-anxiety item**. These are not equivalent targets. A model that predicts “felt nervousness right now” is not directly comparable to one that screens for generalized anxiety disorder or predicts remission in severe MDD. Any cross-study comparison that ignores this is overstating what the literature shows. ⁴⁵

A third issue is **temporal inconsistency**. Different studies use very different units of analysis: framewise AUs, entropy over full interviews, every-100-frame aggregation, 10-second windows, or unspecified summary statistics. This matters because “expressivity” can mean at least four different things: amplitude, frequency, variability, or transition dynamics. The repo currently summarizes mainly **mean and standard deviation**, which is defensible for a baseline but narrower than the best biomarker literature, where temporal variability itself is often one of the most informative signals. ⁴⁶

A fourth issue is **ecological validity versus control**. Controlled tasks can isolate diagnostic signal but may underrepresent real-world behavior. Free-speech post-trauma interviews and clinical conversations are more naturalistic and probably more clinically useful, but they also introduce strong confounds from speaking load, topic content, and interviewer effects. The best path is not to choose one or the other, but to report both. The repo’s split-by-speaking logic is therefore not just an engineering detail; it aligns with an important clinical-methodological principle, although the current mouth-only proxy should be improved. ⁴⁷

Finally, **external validation remains rare**. Reviews of both automated depression diagnosis and PTSD machine-learning studies repeatedly note limited replication, overreliance on a few benchmark datasets, and underreporting of generalization across sites, devices, and populations. This means that very high reported accuracies—especially on private datasets—should be interpreted as encouraging but not yet definitive. ⁴⁸

Recommendations for feature engineering, multimodal fusion, and implementation in the repo

The most important design choice for this repository is to treat emotional expressivity as a **hierarchical phenotype**, not a single score. At minimum, the next version should represent four layers simultaneously: **state** (framewise AUs/emotions), **behavioral episodes** (smiles, brow tension, gaze aversion, blink bursts), **temporal dynamics** (variability, entropy, autocorrelation, onset/offset), and **context** (speaking, listening, question type, valence of prompt). That structure matches the strongest reviewed work much better than a one-pass mean/std summary. ⁴⁹

Recommended feature additions for the repo

The table below is a concrete feature roadmap mapped to the current repo.

Repo area	Current state in inspected code/docs	Recommended change	Why it matters
<code>emotional_expressivity</code> extractor	py-feat emotions + AU columns + <code>mouth_openness</code> ; no explicit gaze/head-pose output in the current summary path. ⁶	Add head pose, eye gaze, blink rate/duration, AU asymmetry , and per-frame confidence/quality flags .	Depression, PTSD, and anxiety studies repeatedly benefit from pose/gaze plus AUs; anxiety especially gains from pose and eye-region cues. ⁵⁰
Speaking split	Speaking probability inferred from rolling std of <code>mouth_openness</code> with 2-component GMM. ⁵¹	Replace or complement with audio VAD + active speaker detection , then keep mouth-based cues as a secondary feature.	Mouth motion is a useful proxy but confounds speech with affective mouth behavior. Free-speech studies are strongest when multimodal. ⁵²

Repo area	Current state in inspected code/docs	Recommended change	Why it matters
Baseline normalization	Baseline video mean division with +1 offset. ⁶	Keep baseline option, but add robust within-subject z-scoring using median/IQR and optionally a neutral-question baseline.	Helps with sex, age, and subject-specific expressivity differences; literature repeatedly shows subgroup effects. ⁵³
Summary statistics	Mean, standard deviation, overall expressivity. ⁵¹	Add percent time active, episode counts, onset/offset latency, lag-1 autocorrelation, sample entropy, spectral entropy, burstiness, transition statistics.	Harati-type and PTSD-dynamics work imply that variability and flexibility are not optional extras; they are biomarker candidates. ⁵⁴
Context handling	Speaking vs non-speaking only. ⁵⁵	Add question/topic/context labels when available and compute context-specific summaries.	PTSD child work explicitly shows context-dependent classification value, and post-trauma free speech is highly context sensitive. ⁵⁶
Quality control	Face detection failures are converted to NaNs; docs warn about poor video conditions. ⁵⁷	Emit tracking failure rate, occlusion proportion, pose-angle thresholds, illumination score, usable-window percentage.	Clinical deployment needs uncertainty and traceability, not just predictions. ⁵⁸

On the computer-vision stack, the simplest high-confidence option is to continue using **py-feat/OpenFace-style interpretable features** as the primary representation and only use deep networks **after** feature extraction, not instead of it. That preserves interpretability and privacy, aligns with E-DAIC and child-PTSD feature releases, and avoids the brittleness of end-to-end raw-video modeling on small datasets. A practical stack would be: **OpenFace or py-feat for AUs/pose/gaze, MediaPipe Face Landmarker/Iris** as a fallback or cross-check, **RetinaFace** or equivalent for robust tracking, and optical-flow or landmark-velocity descriptors if the videos are high enough quality to justify micro-dynamics. ⁵⁹

For models, the repo should support three tiers rather than one “best” architecture. First, an **interpretable baseline**: logistic regression / elastic net / XGBoost on window-level summaries. Second, a **sequence model**: BiLSTM or 2–4 layer transformer encoder over AU+pose+gaze sequences. Third, a **multimodal fusion model**: cross-attention or gated late fusion across face, audio, and text. The empirical record reviewed here strongly suggests that this ladder is better than jumping directly to a large opaque end-to-end model. ⁶⁰

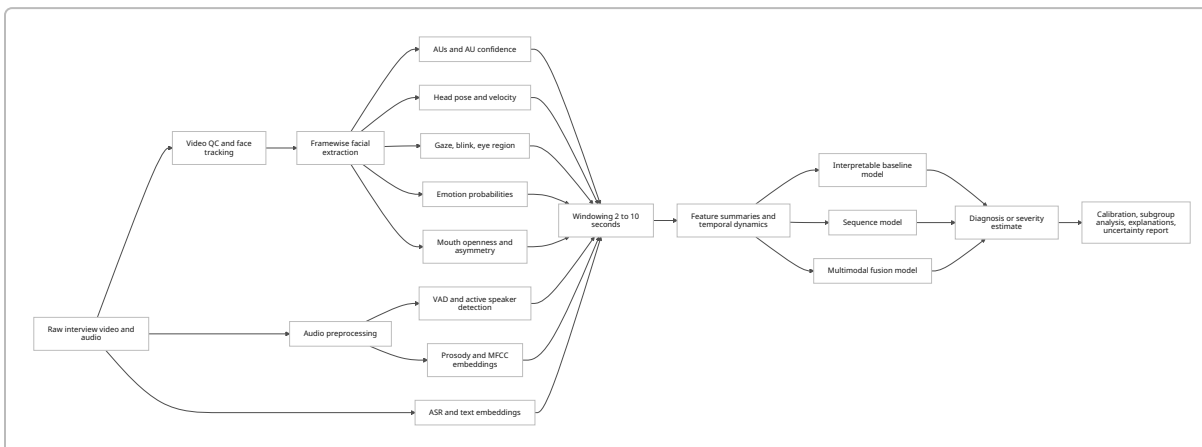
Recommended feature list

This feature set is the most defensible next increment for the current repo:

Feature group	Concrete variables	Priority
Core AUs	AU01, AU02, AU04, AU05, AU06, AU07, AU10, AU12, AU14, AU15, AU17, AU20, AU23, AU24, AU25, AU26, AU45; both intensity and presence	Very high
Valence/arousal proxies	Positive-AU composite, negative-AU composite, neutral proportion, anger/fear/sadness intensity	Very high
Mouth and speech separation	Mouth openness, lip compression, smile amplitude, speaking probability, non-speaking expressivity	Very high
Head dynamics	Mean/SD/range/velocity of yaw, pitch, roll; nodding and shake episode counts	High
Gaze and eyes	Horizontal/vertical gaze, fixation stability, gaze aversion proportion, blink count, blink duration	High
Temporal dynamics	Windowed mean/SD, coefficient of variation, sample entropy, spectral entropy, autocorrelation, burstiness, transition counts	Very high
Symmetry and laterality	Left-right AU asymmetry, unilateral smile/brow activity, head-turn bias	Medium
Quality and confounds	Detection confidence, missingness ratio, occlusion ratio, illumination flag, speaking ratio	Very high

The justification is straightforward: **AUs** are the strongest common denominator across depression and PTSD work; **head pose and gaze** materially improve anxiety-related detection and often help depression as complementary channels; and **temporal dynamics** are where much of the clinically meaningful signal likely resides but are underrepresented in the current repo output. ⁶¹

Proposed experimental pipeline



A concrete training recipe would be: extract framewise features at the native frame rate, then resample to **10 fps** after extraction for efficiency; use **2–10 second windows** with **50% overlap**; compute both raw sequences and window summaries; train **XGBoost** as the first strong baseline, then a **transformer encoder** with hidden size **128–256**, **2–4 layers**, **4–8 heads**, dropout **0.1–0.3**, and class-balanced or focal loss where labels are imbalanced. For XGBoost, a sensible starting space is `max_depth=3–6`, `learning_rate=0.03–0.1`, `subsample=0.7–0.9`, and `colsample_bytree=0.5–0.8`. These are recommendations rather than retrieved source facts, but they are directly aligned with the feature families and dataset sizes represented in the reviewed literature. ⁶²

For evaluation, the repo should standardize on: **subject-exclusive splits**, **nested cross-validation**, **AUC + F1 + balanced accuracy + sensitivity/specificity**, **calibration metrics** such as Brier score or ECE, **bootstrapped confidence intervals**, and **subgroup performance** by sex, age, and site. For clinical tasks, every model should also report whether labels are based on **diagnosis**, **severity scales**, or **state self-report**, because these targets are not interchangeable. ⁶³

Open questions and concluding assessment

Three open questions remain especially important. First, the field still lacks enough studies that report **standardized, directly interpretable effect sizes** for individual facial features rather than only model-level accuracy. Second, the comparative value of **AUs versus head/gaze dynamics** is still under-tested in PTSD, where context seems to matter as much as feature family. Third, the literature remains sparse on **truly external validation across devices, clinics, and demographics**, which is exactly the gap that a careful OpenWillis-aligned benchmarking effort could help close. ⁶⁴

The bottom line is that the repo is already pointed in the right direction: it prefers **interpretable facial phenotypes**, supports **baseline correction**, and tries to control for **speech context**. Those are the correct instincts. The next leap in value will come from broadening the facial phenotype space beyond emotions and mean/std summaries into **AU-centric, gaze/pose-aware, temporally explicit, multimodal digital biomarkers**. If that is done with subject-level validation, subgroup analysis, and uncertainty reporting, this project could move from a useful feature extractor to a genuinely competitive research platform for PTSD, depression, and anxiety detection. ⁶⁵

1 17 24 25 27 30 35 39 40 47 52 <https://www.cambridge.org/core/journals/psychological-medicine/article/deep-learningbased-classification-of-posttraumatic-stress-disorder-and-depression-following-trauma-utilizing-visual-and-auditory-markers-of-arousal-and-mood/733197598EA30BC8379D151173AEFF8F>
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2 4 55 65 https://github.com/Pelmeshek1706/emora/blob/main/emotional_expressivity/emotional_expressivity.md
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9 44 49 54 https://www.researchgate.net/publication/334643130_Classifying_Depression_Severity_in_Recovery_From_Major_Depressive_Disorder_via_Dynamic_Facial_Features
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33 <https://pubmed.ncbi.nlm.nih.gov/40122249/>
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